

BFSS coefficient audit clarification

Technical addendum for review

6 October 2026

1 Explicit slow Clifford contraction and finite sector transport

This supplement expands two compressed steps in Sections 3 and 6.1 of *BFSS leading coefficient cancellation audit*, 6 October 2026. It concerns the uncut leading coefficients at fixed finite N , using the kinetic-one, reduced-kinetic, and frozen-pair definitions of Appendices A, B, and D of the supplied v2.1 manuscript. The identities below do not establish an all-vector reserve, form-domain compatibility, the global Hardy theorem, or constants uniform in N .

1.1 The pair contraction before any slow Clifford trace

Fix an oriented edge with block sizes n, m , put $d = nm$, and choose its spatial frame so that the center difference is reg , $r > 0$. Let t_c be a trace-orthonormal real basis of the block-diagonal slow color space, including the normalized relative center. Its Hermitian action on $\text{Hom}(\mathbb{C}^m, \mathbb{C}^n)$ is

$$T_c q = (t_c)_{aa} q - q(t_c)_{bb}, \quad K_{cc'} = \text{tr}_{\mathbb{C}^d}(T_c T_{c'}).$$

Other blocks can be included in an ambient orthonormal slow basis: their restriction merely adds zero actions and an orthogonal change of the center basis. In particular the following calculation applies to every partition. Since T_c and $T_{c'}$ are Hermitian, K is real symmetric.

Let a, h, k label the $2d$ real off-block color generators. With $[t_A, t_B] = if_{ABC} t_C$, set

$$(F_c)_{ah} = f_{ahc}.$$

The symbol F_c in this subsection denotes a color-action matrix, not the inverse-Gauss coefficient F in Appendix D. On the underlying real space of $\mathbb{C}^d$, F_c is the realification of iT_c . Let J be multiplication by i on that real space. Then

$$\sum_{a,h} (F_c)_{ah} (F_{c'})_{ah} = 2K_{cc'}, \tag{A0a}$$

$$\sum_{a,h,k} (F_c)_{ah} (F_{c'})_{ak} J_{hk} = 0. \tag{A0b}$$

Indeed the first expression is $\text{tr}_{\mathbb{R}}(F_c^T F_{c'})$. For the second, $F_c^T F_{c'}$ is the realification of $T_c T_{c'}$; contracting with J gives, up to the immaterial orientation sign, $2\text{Im tr}_{\mathbb{C}}(T_c T_{c'}) = 0$. This vanishes for each c, c' , not just after summing $c = c'$.

Write $\xi_{h\alpha}$ for fast and $\eta_{c\beta}$ for slow Majoranas, with $\{\xi_i, \xi_j\} = \delta_{ij}$, $\{\eta_i, \eta_j\} = \delta_{ij}$, and $\{\xi_i, \eta_j\} = 0$. The real symmetric $\text{Spin}(9)$ matrices satisfy $\{\Gamma_i, \Gamma_j\} = 2\delta_{ij} I_{16}$. In the stated orientation the centered covariance is

$$\langle \xi_{h\alpha} \xi_{k\gamma} \rangle_f = \frac{1}{2} (\delta_{hk} \delta_{\alpha\gamma} - i J_{hk} (\Gamma_9)_{\alpha\gamma}). \tag{A1}$$

Changing the orientation changes both occurrences of the complex structure consistently and does not affect the results.

The mixed slow-fast part of the off-block Gauss spin is

$$G_a = -i \sum_{h,c,\beta} (F_c)_{ah} \xi_{h\beta} \eta_{c\beta}.$$

Move every slow generator past the second fast generator before taking the fast expectation. The minus sign from this move cancels $(-i)^2$. Equations (A0a)–(A1) give the following equality in the *untraced* slow Clifford algebra:

$$\sum_a \langle G_a^2 \rangle_f = \sum_{c,c',\beta,\gamma} Q_{c\beta,c'\gamma}^G \eta_{c\beta} \eta_{c'\gamma}, \quad Q_{c\beta,c'\gamma}^G = K_{cc'} \delta_{\beta\gamma}. \quad (\text{A2})$$

There is no residual polarized bilinear: its color coefficient is exactly (A0b). Applying CAR to the real symmetric coefficient in (A2) gives

$$\sum_a \langle G_a^2 \rangle_f = 8 \sum_c K_{cc} I = 8d(n+m)I. \quad (\text{A3})$$

The last equality follows from the trace-normalized endpoint Casimirs:

$$\sum_{c \text{ internal}} K_{cc} = (d-1)(n+m), \quad K_{\text{relative center, relative center}} = n+m.$$

The centered inverse-Gauss square multiplies (A3) by r^{-2} .

For the source, let y_{ia} , $1 \leq i \leq 8$, be the real transverse oscillator coordinates; their covariance is $\langle y_{ia} y_{jb} \rangle = \delta_{ij} \delta_{ab} / (2r)$. In the conventions of the audit the mixed Yukawa source is

$$S_e(0) = \left[-2i \sum_{i=1}^8 \sum_{a,h,c,\alpha,\beta} y_{ia} (\Gamma_i)_{\alpha\beta} (F_c)_{ah} \xi_{h\alpha} \eta_{c\beta} \right] U_e(0). \quad (\text{A4})$$

Here $U_e(0)$ is the normalized bosonic/Fock vacuum embedding; (A4) is a map from the slow module, so $S_e(0)^* S_e(0)$ leaves that module untraced. Moving slow past fast generators as above and contracting y, ξ gives

$$S_e(0)^* S_e(0) = \sum_{c,c',\beta,\gamma} Q_{c\beta,c'\gamma}^Y \eta_{c\beta} \eta_{c'\gamma}, \quad Q_{c\beta,c'\gamma}^Y = \frac{16}{r} K_{cc'} \delta_{\beta\gamma}. \quad (\text{A5})$$

For clarity, the unpolarized coefficient is $4(2r)^{-1} \cdot \frac{1}{2} \cdot 2K_{cc'}$ times $\sum_{i=1}^8 \Gamma_i^T \Gamma_i = 8I$. The polarized coefficient contains $\sum_{i=1}^8 \Gamma_i^T \Gamma_9 \Gamma_i = -8\Gamma_9$ but vanishes by (A0b). Consequently CAR yields the exact operator identity

$$S_e(0)^* S_e(0) = \frac{128d(n+m)}{r} I. \quad (\text{A6})$$

The source has one bosonic and one fermionic excitation on this edge, each of energy $2r$. Thus its centered resolvent contraction equals $32d(n+m)r^{-2}I$, as claimed in Section 3 of the audit. This calculation establishes the slow-Clifford reduction directly; a trace over the slow module would not suffice to establish it.

1.2 Transport preserving actual excitation sectors

Fix the centers and use $A(t) = tA$, $0 \leq t \leq 1$. On each edge, let the real positive kinetic and potential matrices be $G_e(t)$ and $V_e(t)$. Set

$$\Omega_e = G_e^{-1/2}(G_e^{1/2}V_eG_e^{1/2})^{1/2}G_e^{-1/2}, \quad \Omega_e G_e \Omega_e = V_e.$$

All these expressions concern finite coefficient matrices. Define the unitary map from the fixed dimensionless oscillator space to the actual bosonic fiber by

$$(\mathcal{U}_e \psi)(Y_e) = (\det \Omega_e)^{1/4} \psi(\Omega_e^{1/2} Y_e).$$

Writing $x_e = \Omega_e^{1/2} Y_e$ and $M_e = \Omega_e^{1/2} G_e \Omega_e^{1/2}$ gives exactly

$$\begin{aligned} \mathcal{U}_e^* H_{B,e} \mathcal{U}_e &= p_{x_e}^T M_e p_{x_e} + x_e^T M_e x_e, \\ \mathcal{U}_e^* (H_{B,e} - E_{B,e}) \mathcal{U}_e &= 2 \sum_{j,k} a_j^* (M_e)_{jk} a_k, \end{aligned} \quad E_{B,e} = \text{tr}_{\mathbb{R}} M_e, \quad (\text{A7})$$

where $a_j = (x_j + \partial_{x_j})/\sqrt{2}$. Thus total bosonic excitation number is invariant even when G_e, V_e do not commute. There is no choice of eigenvectors or division by differences of internal frequencies. At $t = 0$, $M_e = r_e I$.

For fermions let $D_e(t)$ be the Hermitian complex bifundamental mass and put

$$P_e(t) = 1_{(-\infty, 0)}(D_e(t)).$$

With the external gap retained, solve the finite-dimensional unitary transport equation

$$\dot{V}_e^F = [\dot{P}_e, P_e] V_e^F, \quad V_e^F(0) = I. \quad (\text{A8})$$

Then $P_e(t) = V_e^F(t) P_e(0) V_e^F(t)^*$ and $\tilde{D}_e = V_e^{F*} D_e V_e^F$ commutes with $P_e(0)$. Second quantization transports the occupied determinant to the fixed reference determinant. Relative to particles in $1 - P_e(0)$ and holes in $P_e(0)$, the positive excitation matrix in the kinetic-one normalization is

$$B_e^F = 2 \left(\tilde{D}_e|_{1-P_e(0)} \oplus [-\tilde{D}_e|_{P_e(0)}]^T \right). \quad (\text{A9})$$

The transpose acts on the dual hole space. The shifted fermionic Hamiltonian is its fermionic second quantization and preserves total particle-plus-hole excitation number. Its centered value is $2r_e I$ on the one-excitation space.

Uniform pair functional calculus on the normalized thinness ball gives

$$\|M_e - r_e I\| \leq C_N \kappa r_e, \quad M_e \geq c r_e I, \quad \|B_e^F - 2r_e I\| \leq C_N \kappa r_e, \quad B_e^F \geq c r_e I. \quad (\text{A10})$$

These bounds use the external positive/negative gap of D_e and the positive bosonic gap; no internal eigenvalue gap is assumed. The transport is pairwise. In particular its bound never involves an unrelated r_f/r_e .

On the fixed tensor product, let N_e be total bosonic plus fermionic excitation number on edge e . The actual shifted fast Hamiltonian is the sum of the transported pair excitation Hamiltonians. Each spectator edge has zero energy on its transported vacuum because its own ground energy has been subtracted. Hence the spaces

$$\mathcal{E}_e = \{N_e = 2, N_h = 0 \ (h \neq e)\}, \quad \mathcal{E}_t = \{N_e = N_f = N_g = 1, N_h = 0 \ (h \notin t)\}$$

are finite-dimensional, mutually orthogonal, and invariant. The pair source lies in the smaller one-boson/one-fermion subspace of $\mathcal{E}_e$. The monomial inventory in Appendix D(2.6) places

each triangle source in $\mathcal{E}_t$. Different such monomials can mix *within* a sector; the argument neither discards that mixing nor requires individual internal eigenmodes to remain fixed.

Since at most two quanta occur on an edge sector and one on each edge of a triangle, (A10) proves

$$\begin{aligned} H_{e,A} &\geq cr_e I, & \|H_{e,A} - H_{e,0}\| &\leq C_N \kappa r_e, \\ H_{t,A} &\geq cs_t I, & \|H_{t,A} - H_{t,0}\| &\leq C_N \kappa s_t, \end{aligned} \quad s_t = \sum_{h \in t} r_h. \quad (\text{A11})$$

Here all differences are taken after the specified pair transports. The resolvent identity now gives $C_N \kappa / r_e$ and $C_N \kappa / s_t$, respectively, for the inverse differences. Combining these with the actual transported source difference bounds in Section 6.1 of the audit proves its displayed Schur-complement difference estimate. In particular the sources have not been defined by transporting their centered values: one first forms $H_1(A)U(A)$ and then uses the transports to compare it with the centered source.

1.3 Derivative scope and reproducibility

The additional derivatives used for the multiplier in Section 6 of the audit are derivatives of the finite coefficient matrices and their spatial/internal jets. An internal derivative brings its incident r_e^{-1} . A covariant center derivative differentiates r_e , the unit direction, and the normalized internal matrix, again with an incident inverse gap. Along tA , each normalized jet changes by $O(C_N \kappa)$, using one further bounded derivative of the same matrix functions. This explains why coincident internal eigenvalues cause no singularity. The finite root-path inventory and its triangle inequalities remain necessary; smoothness by itself would not control unrelated gap ratios.

The supplementary `check_pair_clifford.py` constructs the untraced slow quadratic coefficient matrices in (A2) and (A5) for $(n, m) = (1, 1), (2, 1), (2, 2), (3, 1)$ at $r = 1$. It compares every coefficient with the displayed Gram-matrix formulas and tests the vanishing antisymmetric coefficients, not just the resulting scalar trace. It uses floating-point arithmetic and reuses the original test's `Spin(9)` and color-basis constructors, with separate contraction code. These finitely many tests are regression evidence; the arbitrary-block result is the preceding algebraic calculation.

The original symbolic script establishes exact rational reductions of its supplied triangle formulas. The original floating script tests four centered singleton-triangle geometries, reconstructing their triangle Gaussian/Fock contractions while supplying the singleton pair Born-Huang and spin terms as known constants. Neither original script computes the nonzero- A operator, verifies the arbitrary-block slow-Clifford calculation, or tests the required slow derivatives. No numerical nonzero- A test is supplied by this supplement either; (A7)-(A11) are analytic statements under the explicitly stated frozen-pair hypotheses. Nothing in these checks certifies the all-vector or global-theorem steps excluded at the start.